

Week 1 Deliverable: Data Understanding Report



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Team No. 10

1. Understanding the Dataset

The dataset provided for this project captures student engagement, demographic backgrounds, and participation outcomes across multiple learning and professional development opportunities. It initially consists of **8,558 records** and **16 variables**.

The opportunities represented in the dataset are categorized into Internships, Courses, Events, Competitions, and general Engagements. Student participation is recorded holistically, capturing personal information (e.g., Gender, Country, Institution, Major) alongside temporal engagement tracking (e.g., Sign-up Date, Apply Date) and final application outcomes (e.g., Status Description and Status Codes). Understanding these variables is critical for establishing an analytical foundation to uncover patterns in student participation and predict future engagement success.

2. Data Dictionary

The following table documents the variables present in the dataset, their specific data types, and what they represent:

Feature Name	Description	Data Type
Learner SignUp DateTime	The timestamp when the student registered for the platform or opportunity.	Datetime/String
Opportunity Id	A unique alphanumeric identifier for the specific opportunity.	String
Opportunity Name	The formal title of the program, course, event, or internship.	String
Opportunity Category	The classification of the opportunity (e.g., Internship, Course, Event).	String
Opportunity End Date	The official closing date and time for the opportunity.	Datetime/String
First Name	The student's given name.	String
Date of Birth	The student's birth date, which can be used to calculate age.	Date/String
Gender	The gender identity of the student.	String
Country	The student's country of residence or origin.	String
Institution Name	The university or educational institution the student attends.	String
Current/Intended Major	The student's academic field of study.	String
Entry created at	The exact timestamp when the record was generated in the system.	Datetime/String
Status Description	The text description of the student's current progress/outcome (e.g., Started, Rejected, Team Allocated).	String
Status Code	A numerical code corresponding to the Status Description.	Integer
Apply Date	The date and time the student submitted their application.	Datetime/String
Opportunity Start Date	The date and time the opportunity officially commences.	Datetime/String

3. Initial Data Preparation

To ensure the dataset is reliable for exploratory data analysis and future modeling, several data cleaning and formatting steps were performed:

1. **Handling Missing Values (Demographics):** The Institution Name and Current/Intended Major columns each contained 5 missing values. Because this represents less than 0.1% of the total dataset, these rows were dropped to prevent downstream analytical errors.
2. **Handling Missing Values (Temporal):** The Opportunity Start Date column contained 3,794 missing entries (~44% of the data). Dropping these rows would result in severe data loss. Instead, these missing values were imputed with the text "Not Specified" so the remaining valid data in those records could be preserved.
3. **Fixing Formatting Inconsistencies:** During inspection, a formatting error was identified in the Apply Date column. Specifically, 107 records had corrupted timestamps with impossible hour values (e.g., 05/11/2023 1085640:21:29). To standardize this, the corrupted time strings were stripped, and only the valid calendar date portion (e.g., 05/11/2023) was retained.
4. **Data Transformation:** Date of Birth was parsed into a standard Datetime format so that a new Age feature can be easily derived during Exploratory Data Analysis.

After cleaning, the final dataset contains **8,553 clean records** ready for analysis.

Metric	Value
Initial Records	8,566
Rows Removed (Missing Institution/Major)	13
Missing Opportunity Start Dates Imputed	3,794
Corrupted Apply Dates Corrected	107
Date of Birth Records Standardized	8,553
Final Clean Records	8,553

4. Summary Statistics

The following baseline statistics provide an initial overview of participation patterns within the dataset:

Table 1: Participation by Opportunity Category

Internships and Courses make up the vast majority of student engagement, accounting for over 87% of all recorded activities.

Opportunity Category	Count of Participants	Percentage of Total
Internship	5,421	63.3%
Course	2,037	23.8%
Event	545	6.4%
Competition	425	5.0%
Engagement	130	1.5%

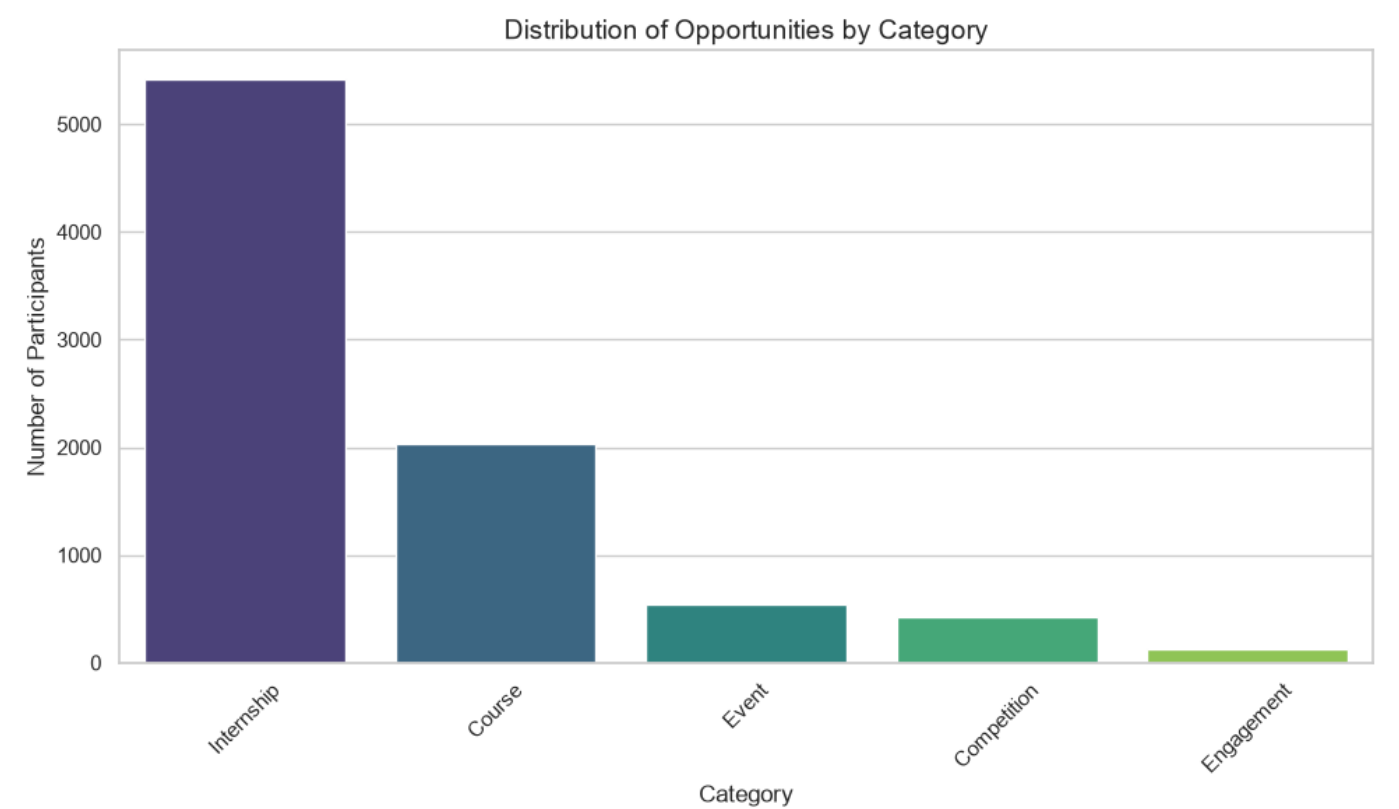


Table 2: Applicant Status Outcomes

While a large portion of students are successfully allocated to teams or start their programs, there is also a high rejection rate, highlighting the competitive nature of these opportunities.

Status Description	Count	Percentage
Rejected	3,569	41.7%
Team Allocated	3,276	38.3%
Started	767	8.9%
Dropped Out	617	7.2%
Waitlisted	109	1.3%
Applied	105	1.2%
Withdraw	86	1.0%
Rewards Award	29	0.3%

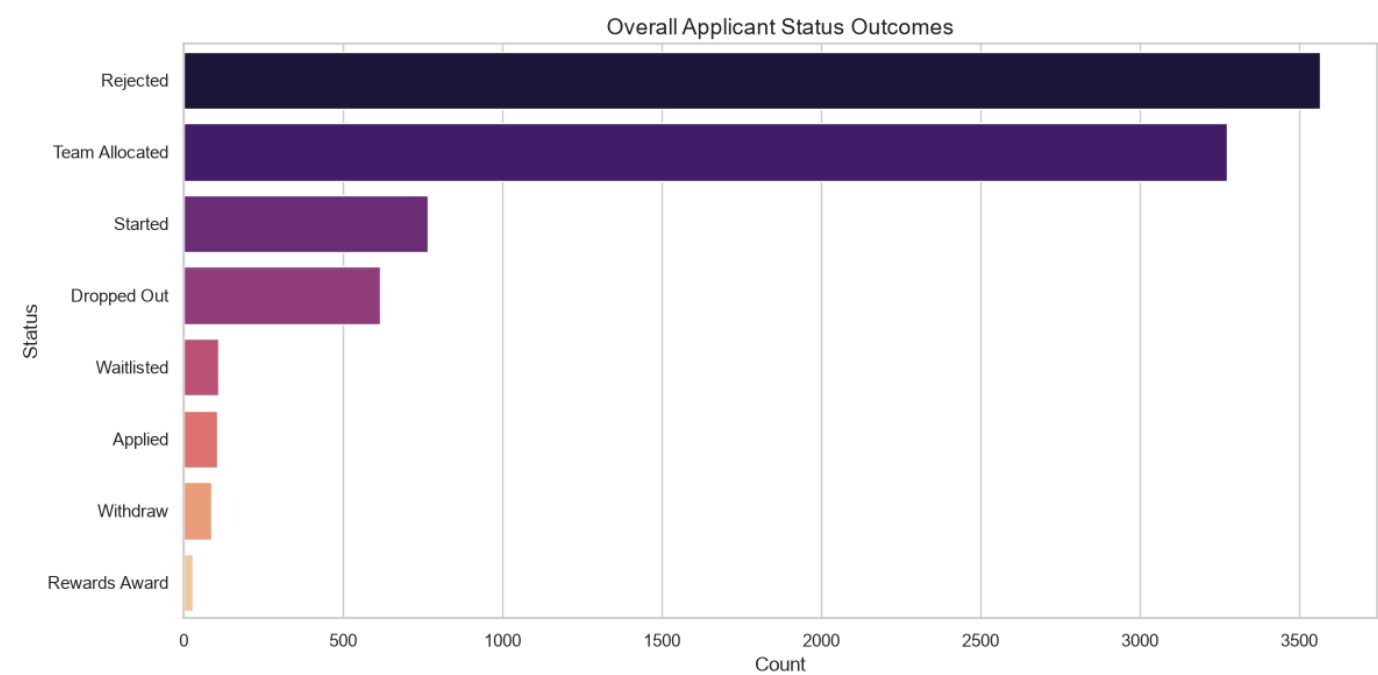


Table 3: Gender Distribution

Participation leans heavily toward male students, making up roughly 59% of the dataset.

Gender	Count	Percentage
Male	5,018	58.6%
Female	3,522	41.2%
Don't want to specify	15	0.1%
Other	3	< 0.1%

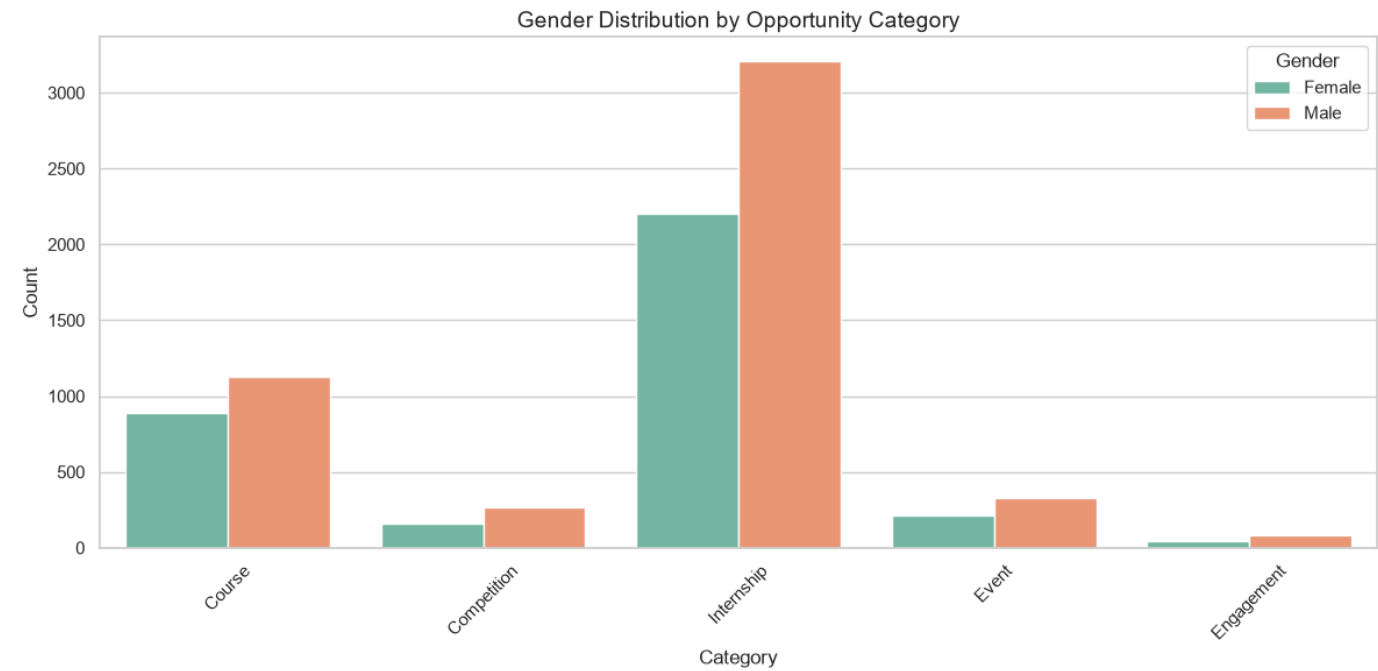


Table 4: Age Distribution of Participants

The participant age distribution is concentrated in the mid-20s, with the majority of respondents falling between 23 and 28 years old. The distribution peaks around age 25–26, indicating that most participants are young adults. A slight right skew is evident, with fewer participants represented in older age groups and a gradual decline in frequency beyond age 30. Very few respondents are younger than 20 or older than 40, suggesting that the sample is predominantly composed of individuals in their early to late twenties.

Statistic	Value
Most Common Age Range	23–28 years
Peak Age	Approximately 25–26 years
Distribution Shape	Unimodal, slightly right-skewed
Majority of Participants	Young adults in their 20s
Age Span Observed	Approximately 16–60 years

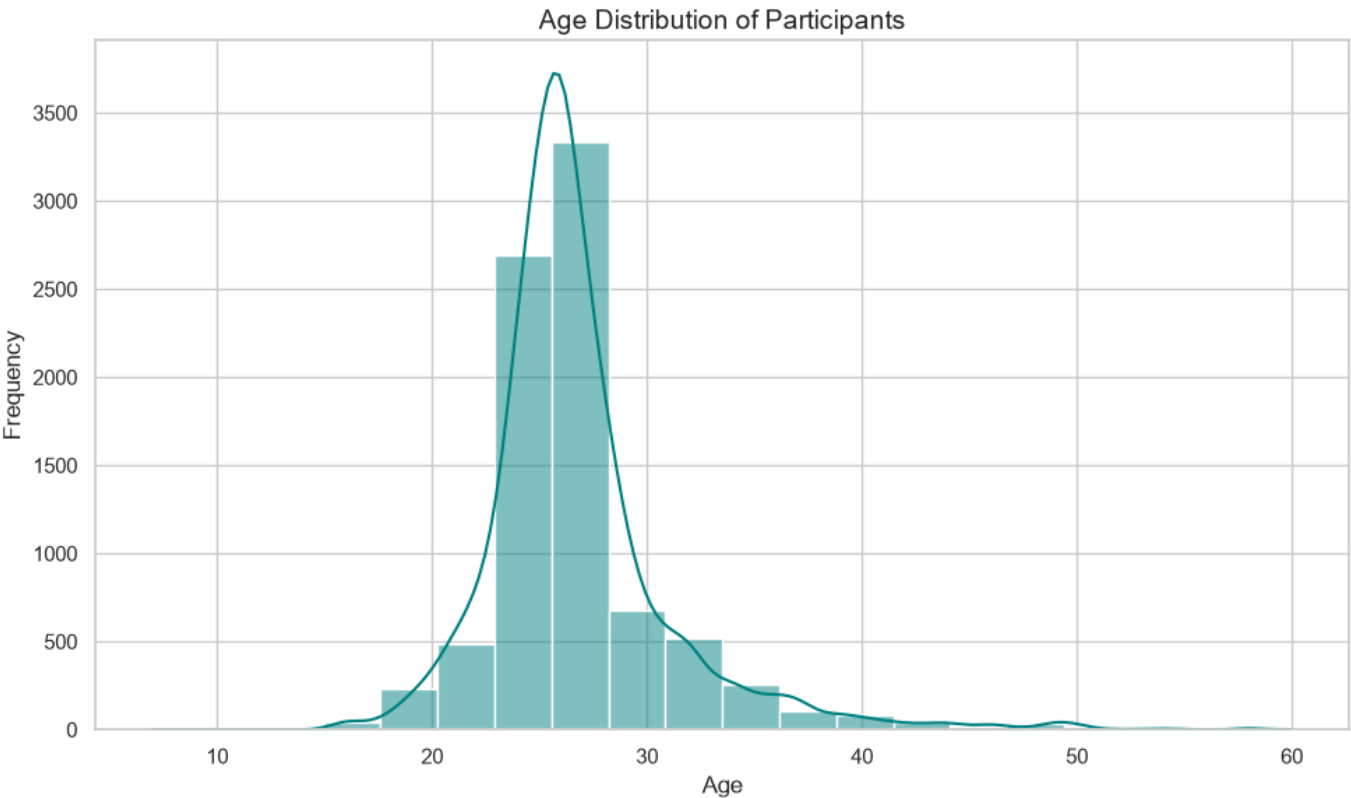
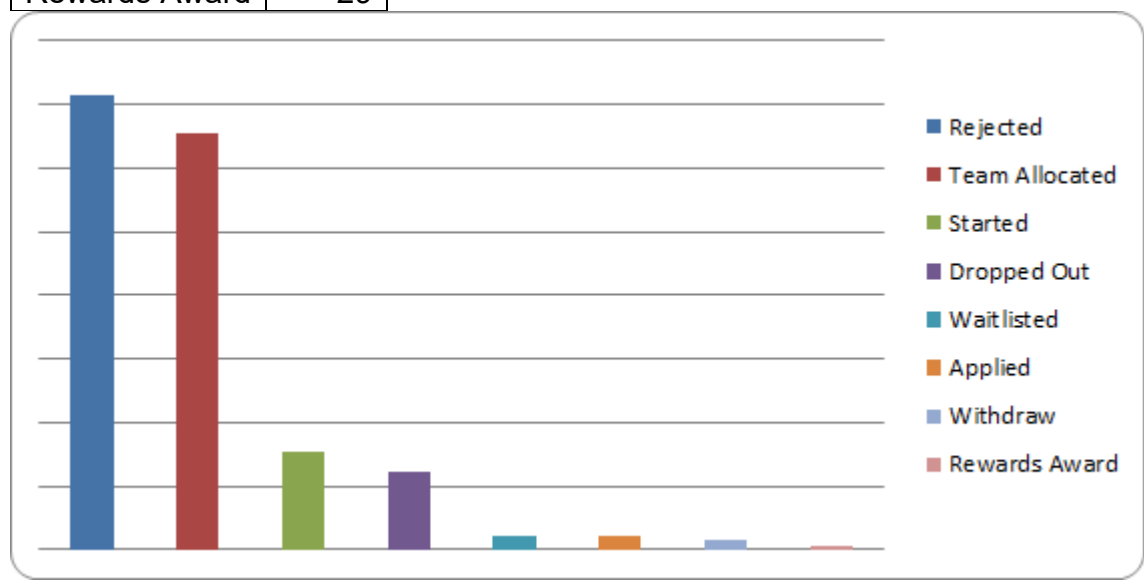


Table 5: Participant Status Distribution

The participant status distribution shows that the largest groups are those who were rejected and those who were allocated to a team, accounting for the majority of records in the dataset. A smaller proportion of participants successfully started their programs, while a notable number dropped out after beginning. Only a limited number of applicants were waitlisted, remained in the applied stage, or withdrew from the process. Very few participants reached the rewards award stage, making it the least common status recorded.

Status	Count
Rejected	3,569
Team Allocated	3,276
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Waitlisted	109
Applied	105
Withdraw	86
Rewards Award	29



Conclusion

The dataset is now thoroughly documented, cleaned, and summarized. The baseline insights reveal that participation is largely concentrated in Internships and Courses, with a predominantly male applicant pool and a highly competitive selection process. These findings establish a solid analytical foundation for the deeper Exploratory Data Analysis (EDA) that will be conducted in Week 2.

Links -

Python Code : <https://drive.google.com/file/d/1y44ggYaDwOdIEQKEV6KIragjIFcaVR7Q/view?usp=sharing>

Excel :

<https://docs.google.com/spreadsheets/d/1QL2oQjytxnxYoDYy8YqOrykww3aox6vf/edit?usp=sharing&oid=104828775998592639386&rtpof=true&sd=true>